PREPARED FOR TIDE

Every school. Every clinic. No one looks you up.

Be reached, not found.

A safe-address layer for family-violence survivors. Services can reach them after they move. No employee, admin or operator can look them up, because the keys live on Tide's Fabric and every access change needs a quorum.

1.5M
Australian women have experienced violence from a previous partner [3]

~2 in 3
NSW women killed by a partner were killed after leaving or while separating [5]

2,215
insider misuse allegations, 7 Queensland agencies, 4 years [7]

0
plaintext addresses stored across 15 Tide operations in the prototype

github.com/BennedictQuanTon/Unfound-Be-reached-Not-found
All people and records in the prototype are synthetic.

built with TideCloak

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1. Executive summary

The problem. When a survivor of family violence leaves, her new address becomes the most dangerous piece of data she owns. Every service she then needs (her child's school, a new GP, the court, the police) stores that address in plain text, readable by staff and by every administrator. The threat is not a hacker. It is one insider with legitimate access doing one lookup. In New South Wales, about two in three women killed by an intimate partner were killed after leaving or while separating [5]. In Queensland, 2,215 allegations of misused confidential information were recorded across seven agencies in four years [7], and a police officer passed a survivor's new address to her abuser [9]. In the UK, four of seven organisations reprimanded over domestic-abuse victims' data in 15 months had revealed a safe address to the alleged abuser [31].

The product. Unfound seals the address once, on the survivor's phone, in three parts: the full street, the school zone and a contact relay. Each service can open only the part it needs. Opening the full address, even in an emergency, takes two of three independent approvers, for one view, and the survivor sees every attempt in her own "Who looked" log.

The boundary. Unfound closes one path: services that hold the address and leak it through a lookup, a permission change or a letter. In a UK survey, 40% of survivors had their new address disclosed to the abuser, mostly by agencies [35]; we estimate Unfound could spare about 1 in 12 survivors that disclosure in a pilot and up to about 1 in 3 if every service joined (section 6.3). It does not stop physical or GPS stalking, people who already know the address, or copies held by services that have not joined. It is one part of a safety plan (section 6.2).

Why Tide. The promise only holds if no one, including the people who run Unfound, holds a key or can grant themselves access. That is Tide's core design. TideCloak encrypts in the browser with keys that exist only as shards across Tide's Fabric. ORK nodes check a role per data tag before they help decrypt. Every access change is a Quorum-Enforced Governance change set signed by a threshold of admins [20][21][22].

What exists. A working prototype with seven personas on three devices, and a 68-second auto-played story running 15 Tide operations with 0 plaintext addresses stored. It also has a live TideCloak code path (DPoP, `doEncrypt` / `doDecrypt`, AdminAPI change sets), a one-command local setup, a film and a cited research base. It was built in four days for RMIT Hackathon 2026, Challenge 2 "UNLOCKED".

The ask. Help to move from prototype to a supervised pilot:

- a hosted TideCloak realm;
- answers to four specific integration questions (section 16);
- access to the Launch Program;
- an introduction to one family-violence service willing to pilot.

2. What Unfound is

One line	A safe-address layer: services can reach a survivor after she moves, and no single employee, admin or operator can look her up.
For whom	Primary: people who have left a violent partner and moved. Secondary: the services that must contact them.
Category	Privacy infrastructure for public and community services (B2G / B2B), free for survivors.
Core idea	One address, three sealed parts, one role per part, no master key, no admin override.
Built on	TideCloak (Keycloak-compatible IAM with BYOiD sign-in), Tide's Cybersecurity Fabric, Hermetic E2EE, Quorum-Enforced Governance.
Stage	Prototype (October 2026). The live Tide path is wired; the public demo runs on a mock fabric that applies the same rules.

What changes for each person

WHO	TODAY THEY SEE	WITH UNFOUND THEY SEE
School	Full address	"In Footscray North zone ✓"
GP clinic	Address and phone	Suburb and a relay number
Court mail	Address in the clerk's file	The street, opened only as the label prints
Police	Any record, any time	Break-glass: 2 of 3 approvals, one view, then revoked
IT admin, DBA, operator	Everything	Ciphertext. A self-grant stays pending.
The survivor	Nothing about who looked	Every attempt, approval, refusal and view

3. Brand

ITEM	VALUE
Name	**Unfound**, written as the lowercase wordmark `unfound`. Two syllables; says what the product protects.
Slogan	**Be reached, not found.**
Support line	Your address, held by no one.
Hero line	Every school. Every clinic. No one looks you up.
Mark	"The pin with no point": a map-pin head drawn as an open arch with a dot inside. The point that would land on a location is missing. Read twice, it is also a doorway.
Palette	Obsidian `#000000`, Canvas `#ffffff`, Fog `#f9fafb`, Slate Mist `#e5e7eb`, Stone `#c9c9c9`, Graphite `#6e6e6e`; Lam Wash `#e2eeff` for sealed surfaces, Lam Seal `#8dbbff` for the number to remember, Lam `#3d8bff` for the one call to action.
Type	Geist (display, weight 600), Inter (UI), system monospace for ciphertext.
Voice	Short, calm, factual. Say "survivor". Never show violence. Name what people can now do, not the security mechanism. Claim only what the boundary covers: "look up", not "find".
Status language	Form and words, not red and green: **SEALED**, **DENIED**, **PENDING**, **APPROVED**.

The full design system is in `docs/DESIGN.md`.

4. The problem

How to read this section: 4.1 and 4.3 show the scale of the danger around a survivor's address. They do **not** measure how much harm Unfound would prevent. 4.2 shows the specific pathway Unfound targets: services leaking the address they hold.

4.1 Leaving is when the risk peaks

FIGURE	WHAT IT MEANS	SOURCE
27%	of ever-partnered women aged 15–49 worldwide have experienced physical or sexual intimate-partner violence	WHO, 2021 [1]
51,100 · 140 a day	women and girls killed by a partner or family member worldwide in 2023	UN Women and UNODC, 2024 [2]
1.5 million	Australian women have experienced violence from a *previous* cohabiting partner (16%), about nine times the 173,300 who experienced it from a current one	ABS PSS 2021–22 [3]
64% · 866,700	of those women moved out when the relationship ended: each move is a new address in school, GP, bank and electoral records	ABS PSS 2021–22 [3]
~2 in 3	women killed by a male intimate partner in NSW were killed after leaving or while separating	NSW DVDRT 2021–23 [5]
46 · 0.43 per 100,000	female intimate-partner homicide incidents in Australia in 2023–24, the largest one-year rise in over a decade (34 the year before). The 2024–25 report, measured on women aged 18 and over, shows a lower rate of 0.29	AIC, Homicide in Australia 2023–24 and 2024–25 [33] [34]
A$22 billion	estimated cost of violence against women and their children in Australia in 2015–16	KPMG for DSS, 2016 [4]
90.4%	of Vietnamese women abused physically or sexually by a husband did not seek help from formal services; about half had never told anyone. Still the latest national study	National Study on VAW in Viet Nam 2019 [12]

4.2 The pathway Unfound targets: services that leak the address they hold

FIGURE	WHAT IT MEANS	SOURCE
4 of 7	UK organisations reprimanded for breaching domestic-abuse victims' data (June 2022 – Sept 2023) had revealed a safe address to the alleged abuser; the seven included a benefits agency, police, councils and a hospital trust	UK Information Commissioner, Sept 2023 [31]
40% · 52%	of UK survivors had their new address disclosed to the perpetrator, **mostly by agencies such as their child's school or local authorities**; 52% for survivors with children (1,019 survivors, 2019 fieldwork)	Citizens Advice / Opinium, Feb 2020 [35]

FIGURE	WHAT IT MEANS	SOURCE
2021	Services Australia disclosed a survivor's new address to her former partner; the OAIC ordered an apology, A$19,980 compensation and an audit of how contact details are updated for separating couples	OAIC, via The Conversation [36]
2023	an English council saved a survivor's new address in a notes field, then posted a letter containing it to her old home, where her ex-partner opened it	ICO reprimand, Charnwood BC [32]
2,215	allegations of misused confidential information in four years across seven Queensland agencies	CCC Queensland, 2019 [7]
4.8×	growth of those allegations outside the police (health, education, corrections, transport): 65 → 312 a year	CCC Queensland, 2019 [7]
87%	of police allegations (1,391 of 1,591) were unauthorised access or unauthorised disclosure	CCC Queensland, 2019 [7]
2014	a Queensland officer searched the police database and passed a survivor's new address to her abuser; the tribunal found the police service liable	QCAT, reported [9]
"since at least 2017"	Queensland police knew their system auto-filled victims' addresses into documents served on offenders	Qld Information Commissioner, Dec 2025 [10]
US$17.4 million	average annual cost of insider risk per organisation; 81 days to contain an incident	Ponemon / DTEX, 2025 [17]
1,205	data-breach notifications to the OAIC in 2025, an all-time high; health providers were the most affected sector (19%)	OAIC, 2026 [16]

4.3 Outside Unfound's boundary: technology-facilitated stalking

These figures explain why Unfound must sit inside a wider safety plan. Unfound does not detect trackers or shared-location accounts.

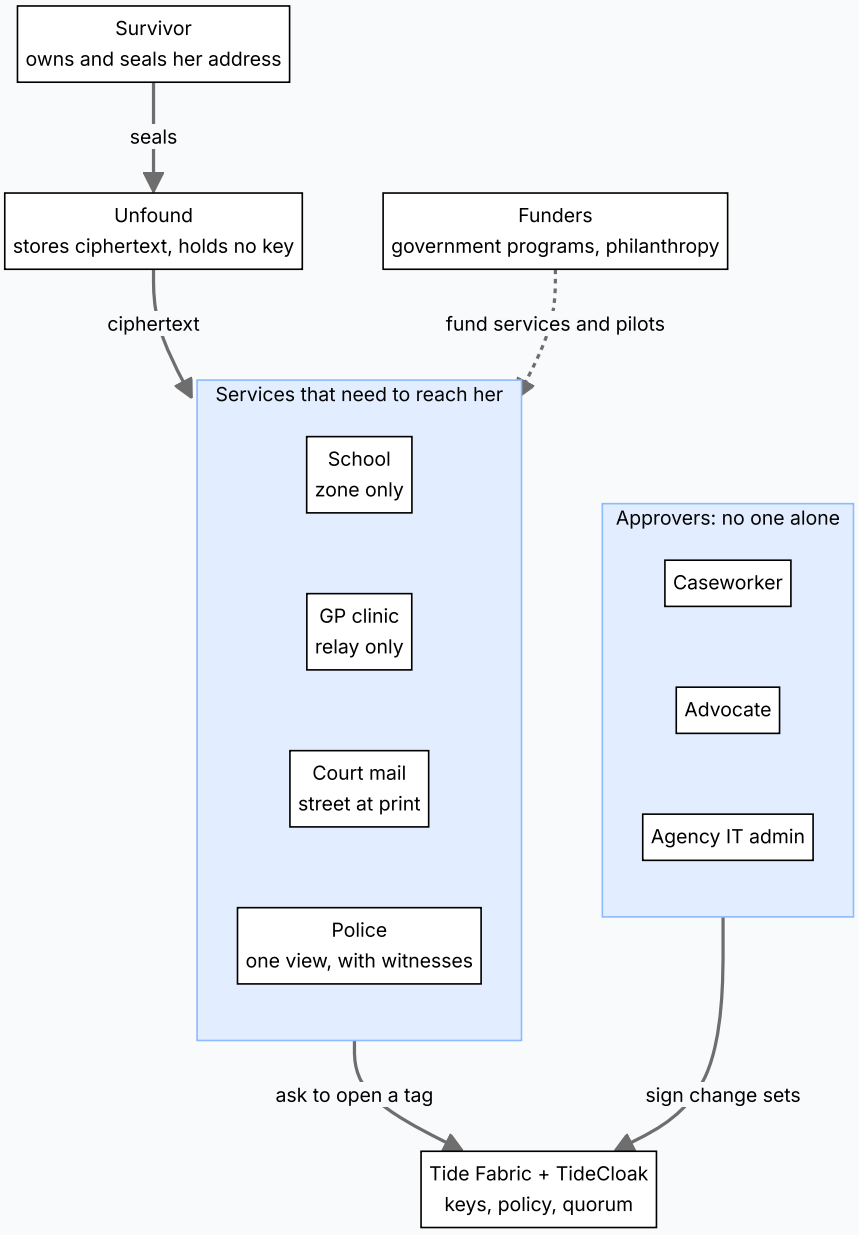
FIGURE	WHAT IT MEANS	SOURCE
99.3%	of family-violence practitioners surveyed had clients who experienced technology-facilitated stalking and abuse	WESNET, 2020 [13]
~1 in 3 · +245%	victim-survivors tracked by GPS, up 245% on the 2015 survey	WESNET via Safe and Equal [13][14]

4.4 Why this stays unsolved

- **Permissions do not fix it.** Role-based access and audit logs are configured and overridden by administrators, the very people in the threat model. Logs show harm only after it has happened.
- **Address programs are narrow.** Australia's silent-elector scheme covers the electoral roll only [24]. In the US, about 39 state Address Confidentiality Programs substitute a mailing address [25][26], but they rely on manual forwarding and on agencies agreeing to accept the substitute. California's program alone had 6,212 active participants in 2023, and more than half were school-aged children [27].

- **Encryption at rest does not fix it.** The provider holds the key, so the operator, and anyone who compromises the operator, can read everything [23].
- **Regulation is tightening.** Australia's statutory tort for serious invasions of privacy commenced on 10 June 2025 [18]. Viet Nam's Personal Data Protection Law (No. 91/2025/QH15) took effect on 1 January 2026, with fines up to 5% of annual revenue for some violations [19]. Agencies now need to *prove* that staff cannot misuse data.

5. Stakeholders and ecosystem



STAKEHOLDER	WHAT THEY NEED	WHAT THEY DO IN UNFOUND	WHAT THEY CAN NEVER DO
Survivor	To use services without creating a new way to be found	Seals her address, chooses services, reads "Who looked"	Lose control of who holds which part
Caseworker (family-violence service)	To onboard a client fast and safely	Seals on the client's behalf, votes on access requests	Read the address
School	Enrolment within catchment, a way to contact the parent	Confirms "in zone"; contacts through the relay	See the street

STAKEHOLDER	WHAT THEY NEED	WHAT THEY DO IN UNFOUND	WHAT THEY CAN NEVER DO
GP clinic	Recall, referrals, a way to contact the patient	Uses suburb and relay number	See the street
Court registry	To serve and post documents	Opens the street only as the label prints	Put the address in a file the other party can read
Police	Welfare checks, protection-order service	Requests break-glass; opens once after 2 of 3 approvals	Browse addresses at will
Independent approvers (advocate, second agency)	Assurance that emergency access is real and necessary	Approve or deny change sets	Approve alone
Agency IT admin	To run systems and pass audits	Operates the realm; one vote among several	Grant themselves access
Unfound (operator)	To run the service without becoming a risk	Stores ciphertext, routes requests	Decrypt anything
Tide	Real-world, high-stakes use of the Fabric	Holds key shards, enforces policy and quorum	Learn what data it protects
Funders and regulators	Evidence of safer outcomes and compliance	Fund pilots; receive aggregate, non-identifying metrics	See individual records

6. The solution

6.1 Five moves

	WHAT HAPPENS	TIDE PRIMITIVE
01 · Seal once	The address is split into `addrfull` (street, unit, postcode), `addrzone` (catchment) and `addrcontact` (suburb, relay number), and each part is encrypted in the survivor's browser.	`doEncrypt` with tags; Hermetic E2EE
02 · Share the minimum	Each service holds a role for exactly one tag.	`_tide_<tag>.selfdecrypt`, checked by ORKs
03 · No one alone	Any role grant is a change set; two of three realm admins must sign it. A lone commit returns `412 QUORUM_NOT_MET`.	Quorum-Enforced Governance, threshold signatures
04 · Emergencies, with witnesses	Police request break-glass; two approvers sign; one view; the role is revoked.	Signed change sets
05 · Survivor sees who looked	Every attempt, approval, refusal and view appears in her own log.	Change-set history, Fabric results

6.2 Scope: what Unfound protects, and what it doesn't

COVERED	NOT COVERED
Lookups by staff or compromised accounts inside a participating service: each opens only the tag its role allows, and every attempt is logged	Being followed, GPS trackers, shared location accounts
An admin granting themselves access: a change set that needs 2 of 3 approvers	Friends, family or neighbours who already know the address
A breach of Unfound's servers or database: ciphertext only	Plain-text copies kept by services that have not joined
Letters and forms produced through Unfound: the street is filled in only at print, never saved into a template, note or export	What an authorised person writes down after a legitimate view
	The survivor's own phone if taken or compromised (reduced by design, not prevented)

Each claim in this brief should name: the participating service and workflow, the disclosure path it interrupts, the control that enforces it, and the exposure that remains.

6.3 How much of the problem this reaches (estimate)

There is no study that splits every way an abuser locates a survivor. The closest measure is the Citizens Advice survey [35]: **40%** of survivors had their new address disclosed to the perpetrator, mostly by agencies. We use it as the size of the path Unfound targets, then apply two assumptions that the pilot must test.

STEP	VALUE	BASIS
A. Survivors exposed through an agency	40% (52% with children)	Survey, self-reported, Great Britain, 2019 [35]
B. Share of agency leaks Unfound's controls can stop inside a joined service	70–90%	Our assessment: lookups, self-grants, phone pretexting (staff cannot read what they cannot see), letters and forms generated through Unfound are stopped; contact-update errors partly; an authorised viewer's notes are not
C. Share of a survivor's services that have joined	25% pilot · 70% regional · 100% ceiling	Assumption: a pilot connects about 3 of the 10–12 services one family uses

SCENARIO	A × B × C	READING
Supervised pilot	40% × 80% × 25% ≈ **8%** (range 6–11%)	About 1 in 12 survivors spared an agency disclosure
Regional rollout	40% × 80% × 70% ≈ **22%** (range 17–29%)	About 1 in 5
Every service joins (ceiling)	40% × 70–90% ≈ **28–36%**	About 1 in 3

In terms of routes, Unfound closes one of five (agencies; friends and family; device and GPS tracking; physical following; public records and social media). It is the one the survey shows most often, and the only one a survivor cannot close herself: she can change her phone or her friendships, but not how a school files a letter.

What these numbers are not: they measure disclosure, not harm prevented. Routes overlap, so the scenarios are upper bounds for that path, and the source is British. Steps B and C become pilot metrics: blocked lookups, letters generated without a stored address, and plain-text copies removed.

6.4 The story the prototype plays (`/demo?story=1`)

SCENE	PERSON · DEVICE	WHAT HAPPENS
1	Linh · iPhone	Seals her new address: three ciphertexts, zero plaintext.
2	Westgate Primary · MacBook	Confirms "Footscray North zone ✓". Trying to open the street: **DENIED** by the network.
3	Dan, IT admin · MacBook	The legacy database shows every address; Unfound shows ciphertext. His self-grant becomes a change set; committing it returns **412**.
4	Const. Reyes · iPad	Requests a welfare check: change set **PENDING (0/2)**.

SCENE	PERSON · DEVICE	WHAT HAPPENS
5–6	Mai and Ruth, approvers	Deny Dan's request; approve Reyes's: quorum met.
7	Const. Reyes · iPad	Opens the address once; the role is revoked automatically.
8	Linh · iPhone	"Who looked": every event, including Dan's attempt.

What we chose **not** to build: server-side decryption, an admin override, plaintext logs, a master key.

7. How Unfound uses Tide

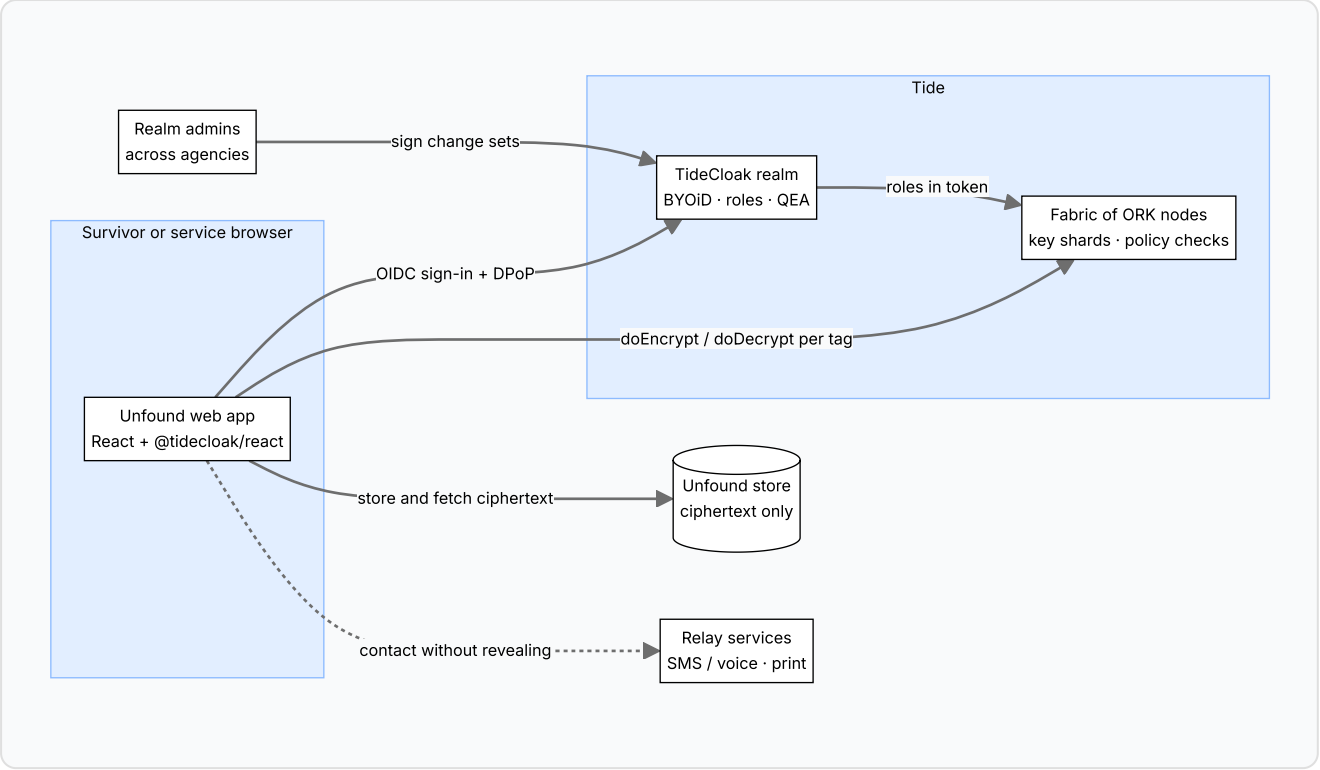
TIDE COMPONENT	WHAT IT IS	WHAT UNFOUND USES IT FOR	WHY IT MATTERS HERE
TideCloak	Keycloak-compatible IAM secured by Tide's Fabric [20]	Realm, users, roles, OIDC sign-in for every persona	Standard integration for agencies already on OIDC
BYOiD	Users bring a Tide identity they control [20]	Survivors and staff sign in without Unfound storing passwords	No credential database to leak
DPoP	Sender-constrained tokens (`useDPoP` , relay page `tide_dpop_auth.html`)	Tokens bound to the browser that requested them	A stolen token does not work on another machine
Cybersecurity Fabric / ORKs	Decentralised nodes holding key fragments; "no single node holds a complete key. Ever." [21]	Holds the keys for every address tag	Nobody at Unfound, the agency or Tide can decrypt alone
Threshold	A minimum number of ORKs must take part, typically 14 of 20 [21]	Every encrypt and decrypt	Compromising one node or one server is not enough
Hermetic E2EE: `doEncrypt` / `doDecrypt`	Encryption in the user's browser with tagged data [21]	Seals the three address parts; opens one part per service	Our server holds ciphertext only
Tag roles: `_tide_<tag>.selfencrypt` / `.selfdecrypt`	Realm roles checked before decryption	One role per tag per service: zone for schools, contact for GPs, full only by quorum	Least privilege is enforced by the network, not by our UI
Quorum-Enforced Governance (QEA)	Admin changes are proposed, reviewed and approved by several admins, signed with threshold signatures, all logged [21][22]	Every role grant, including break-glass; `AdminAPI.addUserRoles` , `getPendingChangeSets` , `commitChangeSet`	Ends "god-mode": an admin cannot grant themselves access
Change-set history	Every proposal, vote and outcome is recorded [21]	Feeds the survivor's "Who looked" log	Turns audit into something the survivor can see
Forseti contracts (planned)	Custom policy validated by the ORKs [21]	One-time views and time-boxed break-glass enforced by the network	Moves "open once" from app logic into the Fabric

TIDE COMPONENT	WHAT IT IS	WHAT UNFOUND USES IT FOR	WHY IT MATTERS HERE
Tide MCP	Tide's agent tools (`tide_hosting` , `tide_doctor`)	Chose local vs hosted realms and checked the integration; declared in `.mcp.json`	Repeatable, checkable setup for the next developer

Code map

FILE	TIDE FEATURE
`src/demo/live.tsx`	`TideCloakContextProvider` with DPoP, sign-in, `doEncrypt` / `doDecrypt` , AdminAPI change sets
`src/demo/engine.ts`	Tags, `_tide_<tag>.*` roles, quorum threshold `max(1, floor(0.7 × N))`
`src/demo/fabric.ts`	The `Fabric` interface implemented by live and mock modes
`tools/tide-setup.mjs`	Copies the realm adapter, creates and assigns the six `_tide_addr*` roles
`scripts/setup-local.sh`	TideCloak in Docker, licensed realm with QEA, enrolment, wiring, app start
`docs/TIDE_MCP.md`	How Tide's MCP was used, with `tide_doctor` results

8. Technical architecture



8.1 Stack today

LAYER	CHOICE
Front end	Vite 5, React 18, TypeScript 5.6, react-router 6, motion 12, Lenis
Identity and crypto	`@tidecloak/react` 0.14.33, DPoP ES256
Local realm	`tideorg/tidecloak-dev` in Docker, Tide realm kit, QEA on
Mock fabric	WebCrypto AES-GCM 256 with the same role and quorum rules, so the app runs without a realm
Hosting	Static build (Vercel or Netlify)

8.2 Data model

OBJECT	CONTENTS	STORED WHERE
Sealed record	Three ciphertexts, each with a tag	Unfound store
Service binding	Which service holds which tag role	TideCloak roles
Change set	Requested role, target, reason, votes, status	TideCloak QEA
Event	Who did what to which record, and the result	Change-set history plus the Unfound event log (no plaintext)

8.3 Threat model

ADVERSARY	OUTCOME
Curious or bribed staff member at a service	Gets only the tag their role allows; the attempt is logged for the survivor.
Realm admin granting themselves access	A pending change set; committing alone returns `412 QUORUM_NOT_MET`.
Unfound operator or a database breach	Ciphertext only; no key on the server to steal or subpoena.
Compromised TideCloak server or one ORK	No usable key: keys exist only as fragments across independent nodes.
Stolen access token	Useless from another device (DPoP).
Abuser coercing the survivor's own device	Out of scope for cryptography. Mitigations: quick exit, no address shown by default, caseworker-assisted recovery (roadmap).

9. What exists today

9.1 Prototype metrics

METRIC	VALUE
Build time	4 days (22–25 October 2026), 21 commits
Source	36 TypeScript/TSX files, 4,721 lines, plus 679 lines of CSS
Personas and screens	7 personas (survivor, school, GP, IT admin, two approvers, police), 5 role screens
Devices simulated	iPhone, MacBook, iPad
Auto-played story	8 scenes, 68 s, 15 Tide operations, **0 plaintext addresses stored**
Data tags and roles	3 tags, 6 `_tide_addr*` roles, quorum 2 of 3
Bundle (gzip)	260 KB landing, 67 KB demo chunk, 3 KB DPoP helper
Film	1:55, 1080p, 27 MB, voice by Kokoro, rendered frame by frame from the real prototype
Research base	36 cited sources in this brief
Tide checks (`tide_doctor`)	DPoP client **PASS**, DPoP relay page **PASS**, secrets hygiene **PASS**; adapter waiting on an enrolled realm
Console errors in end-to-end runs	0

9.2 Status: simulated, implemented, verified

LEVEL	WHAT IT COVERS
Verified on live Tide	A licensed realm with Quorum-Enforced Governance. No live encrypt or decrypt has run yet (see below).
Implemented in code	TideCloak sign-in with DPoP, `doEncrypt` / `doDecrypt` per tag, AdminAPI change sets, the six tag roles, one-command setup.
Simulated in the prototype	The eight-scene story on a mock fabric with the same role and quorum rules; relay numbers, court printing and the one-view revoke.

The realm `unfound1` was provisioned and licensed on local TideCloak with QEA on. Creating the first Tide account then failed on every attempt on 22 October 2026, inside Tide's enclave: "Could not reach enough NEW ORKs (got 3 of 20 required)" and "(got 7 of 20 required)". Earlier licence attempts had returned "Midgard.FinalizeWallet: Not enough orks returned an ok response". The live path switches on as soon as `public/tidecloak.json` holds an enrolled realm. Screenshots are in `docs/assets/`.

10. Landscape and alternatives

	SURVIVOR CHOOSES WHAT EACH SERVICE SEES	INSIDERS AND ADMINS CANNOT READ IT	NO SINGLE PERSON CAN GRANT ACCESS	EMERGENCY ACCESS WITH WITNESSES	SURVIVOR SEES WHO LOOKED	WORKS ACROSS SERVICES
Collecting less: each service keeps only what it needs	◐	◐	–	–	–	–
Silent elector (AU) [24]	–	◐	–	–	–	–
Address Confidentiality Programs (US) [25] [26]	◐	◐	–	◐	–	◐
Role-based access + audit logs [7][8]	–	–	–	◐	–	–
Encryption at rest, provider-held key [23]	–	–	–	–	–	◐
Unfound on Tide [20] [21][22]	●	●	●	●	●	●

● yes · ◐ partly · – no. Our assessment, from the public descriptions cited. Unfound's marks hold only for services that join it.

Collecting less is the simplest alternative and should come first: a school that never stores the street cannot leak it. Unfound makes that minimum consistent across services, and adds what minimisation alone cannot: emergency access with witnesses, no single person able to grant access, and a record the survivor can see.

Our edge is not an encryption library. It is the combination of three things. Least-privilege disclosure per service, enforced outside our control. Governance where no single person, including us, can grant access. And a survivor-facing record of every attempt. Tide is what makes all three possible together.

11. Feasibility

Technically feasible now.

- Tide supplies the hard parts: threshold key management, policy enforcement at the ORKs, quorum governance and DPoP.
- What remains is ordinary application engineering: a ciphertext store, relay integrations and onboarding.
- The live path is already written against `@tidecloak/react`, and only enrolment was blocked.
- TideCloak's free tier covers 100 users [28], enough for a first pilot of about 30 survivors, 10 service staff and 3 approvers.

Operationally feasible.

- Services keep their own systems. They hold a role, open one tag and use a relay.
- Schools and clinics need no new hardware.
- Onboarding goes through caseworkers, who already do safety planning.

Fundable.

- Australia's Leaving Violence Program is funded at A$925.2 million over five years [29], within more than A$3.4 billion committed to the National Plan [29]. That is a pathway for funded delivery partners.
- Privacy reform [18][19] gives agencies a compliance reason to adopt.

Market sized from one segment.

- There are 9,653 schools in Australia [30].
- At a hypothetical A$49 per school per month, schools alone are about A$5.7 million a year.
- Clinics, courts and police are additional.
- The price is a hypothesis to test in the pilot.

Proven pattern. US Address Confidentiality Programs show sustained demand, with California's growing about 12% a year [27]. They also show the gap: manual mail forwarding, no cryptographic control, and nothing for the survivor to see.

12. Path to production

AREA	PROTOTYPE TODAY	PRODUCTION REQUIREMENT
Realm	Local Docker	Hosted TideCloak (Skycloak) in an Australian region, separate staging and production realms
Enrolment	Blocked by ORK availability	Monitored enrolment; retry and status checks agreed with Tide
Cross-user decryption	Tag roles assumed realm-wide	Confirmed pattern with Tide (sealed by the survivor, opened by a service role)
One-time view	Revoked by the app after use	Enforced by a Forseti contract on the ORKs
Storage	Browser state	Managed Postgres with ciphertext only, row-level tenancy, backups
Relay	Simulated	SMS and voice relay numbers; print-and-post relay for courts
Onboarding	Persona picker	Caseworker-assisted onboarding, Tide's in-app profile form, account recovery
Survivor safety UX	Calm defaults	Quick exit, disguised app name option, no notifications on shared devices, multi-language (English, Vietnamese first)
Accessibility	Keyboard and reduced motion	WCAG 2.2 AA audit
Assurance	`tide_doctor` checks	Independent penetration test, Privacy Impact Assessment, ISO 27001 roadmap, IRAP if government-hosted
Governance	3 synthetic admins	Approvers drawn from at least two independent organisations; written break-glass policy
Operations	None	Incident response plan, uptime monitoring including Fabric health, support desk
Data	Synthetic only	Data-sharing agreements with each service; retention and deletion on the survivor's request

13. Roadmap

PHASE	WHEN	GOAL	EXIT CRITERIA
0 · Live on Tide	Nov–Dec 2026	Enrolled hosted realm; live mode end to end; Tide's answers on cross-user decryption and one-time views	Full story runs on live Tide; `tide_doctor` all PASS
1 · Design partners	Jan–Mar 2027	Co-design with one family-violence service, one school and one GP clinic; safety review with practitioners	Signed pilot agreement; PIA complete; penetration test passed
2 · Supervised pilot	Apr–Sep 2027	20–30 survivors, 3 services, 3 independent approvers	0 plaintext addresses; 100% of access attempts visible to survivors; median onboarding under 15 minutes; service staff satisfaction ≥ 4/5
3 · Integrations	Oct 2027 – Mar 2028	Connectors for school enrolment and GP practice systems; court print relay; Forseti one-time views	Two system integrations live; break-glass under 10 minutes end to end
4 · Scale	2028	Multi-state rollout in Australia; Viet Nam pilot with a local partner under the PDPL	Funded contracts in two states; first Viet Nam partner

14. Business model

ELEMENT	APPROACH
Survivors	Always free
Paying customers	Agencies and services: schools, clinics, courts, police, family-violence services
Pricing (hypothesis)	Per site per month, tiered by service type; break-glass included
Funders	Government family-violence programs, privacy and cyber resilience budgets, philanthropy
Cost drivers	TideCloak Premium, hosting, relay telephony and print, support
Moat	Network effect across services in one region, trust from practitioners, and Tide's governance model, which a competitor cannot copy with permissions alone

15. Risks and mitigations

RISK	MITIGATION
Fabric availability, as seen on 22 Oct 2026	Health monitoring, graceful degradation (relay contact still works), an agreed incident channel with Tide
Cross-user decryption semantics differ from our assumption	Confirm with Tide in Phase 0; fall back to a Forseti policy or per-service re-encryption by the survivor's client
Survivor's device is compromised or coerced	Quick exit, nothing shown by default, caseworker recovery, safety planning
Services resist new workflows	Start with a relay and "in zone" answers, which are less work than today's process
Approver availability in emergencies	Rosters across agencies, quorum sized to on-call capacity, escalation path to existing police powers
Misreading statistics	Every figure cited with its denominator; Operation Impala counts allegations, not proven cases; population statistics are never presented as harm Unfound prevents
Overclaiming protection	Copy says "look up", not "find"; a published Covered / Not covered boundary; status split into simulated, implemented and verified
Plain-text copies outside Unfound	Map each pilot workflow end to end (collection, viewing, export, correspondence, retention, deletion) and remove or replace each copy before go-live
Ethics	Synthetic data only until approved research ethics and practitioner oversight are in place

16. What we are asking Tide for

1. **A hosted realm.** A Skycloak-hosted TideCloak realm in an Australian region for Phase 0 and the pilot, and access to the Launch Program and Premium tier.
2. **Four technical answers.**
   - Can data sealed by one user with `doEncrypt` on tag *X* be opened by another user who holds `_tide_X.selfdecrypt`? If not, what is the recommended pattern?
   - Can a Forseti contract enforce "one decryption, then expire" for break-glass?
   - Can a realm's QEA quorum include admins from different organisations?
   - Can change-set history be exported per record to power a survivor-facing log?
3. **Enrolment reliability.** What caused the ORK failures on 22 October 2026, and which health signals we should monitor.
4. **Introductions.** One family-violence service and one public agency interested in a supervised pilot.
5. **A case study.** Publish Unfound as a Tide reference build once the pilot is live.

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All people, addresses and records in the prototype are synthetic. Figures from secondary reporting are marked as such; check the primary documents before quoting them publicly.